

MARTIN WARD

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IT INFRASTRUCTURE OPERATIONS | NETWORKS | AI SYSTEMS

SECRET (LEVEL 2) CLEARANCE — OBTAINED APRIL 2026

EDUCATION

Bachelor of Computer Science, Dalhousie University	2021 – 2026
3D Modeling & Visual Effects Production, Humber College	2018 – 2019
Bachelor of Business Administration, Memorial University	2008 – 2013

WORK EXPERIENCE

Programmer (Co-op), Network Communications May 2024 – May 2026

Department of National Defence, CFB Halifax — Base Information Services

- Worked with Cisco, Juniper, and Palo Alto network equipment in a production environment; familiar with Cisco IOS and embedded event manager (EEM).
- Set up and maintained a network traffic capture and analysis system (Arkime) connected to a searchable database (OpenSearch) to store, index, and query large volumes of network packet data for troubleshooting and security investigations.
- Configured and operated network security monitoring tools: Suricata (signature-based intrusion detection for known threats), Zeek (network traffic analysis and behavioural monitoring), and Kitsune (machine learning-based anomaly detection).
- Used network monitoring software (Zabbix) to track IT infrastructure health.
- Familiar with network traffic capture methods including network taps, Wireshark, and tshark for capturing and filtering traffic to support network troubleshooting.
- Wrote PowerShell scripts to allow users to manipulate and filter network traffic data (PCAPs) for analysis and reporting.
- Built an automated documentation platform (MkDocs with CI/CD pipeline) to produce and maintain accurate system documentation, reports, and technical procedures.
- Installed and configured an offline AI processing system (Whisper for speech-to-text, locally hosted language models) to transcribe and summarize audio in an air-gapped network environment.
- Researched and applied AI-enabled tools to support IT infrastructure operations.

Quantitative Modeler (Co-op) Fall 2023

Defence R&D Canada Atlantic (DRDC)

- Researched the operational role of negative information in maritime sensor models to predict adversary intent.
- Implemented geospatial probability algorithms in Python using NumPy, Pandas, GeoPandas, Shapely, and Matplotlib.
- Rendered animated visualizations with FFmpeg used in national defence strategy development.

Fisherman 2021

Qikiqtaaluk Fisheries Corporation — St. John's, NL

- Sustained 12+ hour shifts in demanding North Atlantic conditions during month-long offshore rotations.
- Streamlined onboard packaging and processing workflows by identifying bottlenecks, improving throughput and quality.
- Collaborated with a tight-knit crew to record accurate catch data and ensure full compliance with DFO regulations.
- Built strong physical and mental endurance, situational awareness, and teamwork in a high-stakes environment.

TECHNICAL PROJECTS

Smart Note Scanner & OCR Pipeline

Python, OpenCV, NumPy, Tesseract OCR, Image Processing

- Built an automated pipeline that scans handwritten notes, detects page edges, corrects perspective distortion, and converts text to digital format using optical character recognition.

Security & Identity Sandbox

Number Theory, Bijective Mappings, One-way Functions, Modular Arithmetic

- Built a diagnostic tool that demonstrates the difference between reversible encoding and cryptographic one-way hashing, visualizing how small input changes produce completely different outputs.

Relational Database Architect — Library System

SQL, Normalization (1NF-BCNF), Schema Design, Relational Algebra

- Designed a normalized relational database for a library that efficiently stores books, authors, and members while eliminating duplicate data and maintaining data integrity.

Secure Artist Portfolio Platform

PHP, MySQL, HTML, CSS, PDO Prepared Statements, Session Security

- Built a full-stack web platform with secure user sessions, protected file uploads, and password hashing to prevent common web vulnerabilities.

Pathfinding Visualizer & Algorithm Comparison

Priority Queues, Graphs, Dijkstra, A, BFS, Greedy Best-First*

- Built an interactive tool that visualizes how different search algorithms find the shortest path through a grid, comparing their efficiency and behaviour.

4-Bit Calculator (ALU)

Combinational Logic, Two's-Complement, Multiplexing, OpCode Decoding

- Designed a hardware-level calculator using logic gates that performs addition, subtraction, and logic operations controlled by operation codes routed through multiplexers.

Software-Defined Wireless Mesh Simulator

Python, Distributed Systems, Ad-hoc Routing, Signal Modelling

- Built a simulated wireless network where each device can route data to others without a central hub, using signal strength to calculate connection quality and self-healing when connections fail.

Micro-Bash Custom Command-Line Interpreter

C, POSIX System Calls, File Descriptors, IPC, Signal Handling

- Built a functional Unix shell that launches programs, manages running processes, handles user interruptions, and redirects input and output between programs and files.

Engagement-Aware PageRank Engine

Python, NumPy, Matplotlib, Linear Algebra, Stochastic Processes

- Developed a search ranking algorithm that scores pages based on both link structure and user engagement metrics using probability modelling.

DNA Sequence Aligner — Bioinformatics Tool

Python, Dynamic Programming, Memoization, Big-O Analysis

- Built a tool that compares DNA sequences and calculates the minimum steps needed to transform one sequence into another, used for genetic analysis and research.

Similarity Engine — k-Nearest Neighbors

Metric Spaces, Euclidean Distance, Vector Representation, Min-Max Scaling

- Built a recommendation system that finds similar items by measuring distance between their characteristics in multi-dimensional space.

3D Submarine Simulator & Boids Ecosystem

C/C++, OpenGL/GLUT, GPU Shaders, Boids Algorithm, Rigid-body Dynamics

- Created an interactive 3D environment with autonomous fish that flock together using simple rules, rendered with graphics hardware acceleration.